

The Toll of Homeowners Insurance Non-Renewals on Local Economies

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Abstract: Research shows that homeowners insurance premium increases (home foreclosures) adversely affect household finances (local economic variables). Motivated by the increasing difficulty of obtaining homeowners insurance in climate-risky areas, we examine effects of insurer-initiated non-renewals of homeowners insurance policies using county-year non-renewals during 2018–2023 from the U.S. Senate Budget Committee. We expect non-renewals have more economically and socially adverse effects than premium increases because non-renewals suggest insurers cannot adequately evaluate, manage, or price risks, and because mortgage contracts require hazard insurance on the mortgaged property. We first show that non-renewals are positively associated with three-year average climate-disaster damages in that county and in other counties in the state, with these associations concentrated in states served by financially weaker insurers, that require pre-approval of insurance rates, and that experience more uncertain climate-disaster losses, as well as in counties with more socially vulnerable populations. We then estimate the effects of non-renewals on foreclosures and local economic variables using a shift-share-style instrument for non-renewals and find that instrumented non-renewals are significantly positively associated with foreclosure rates, with this association concentrated in states with fewer insurers, no insurer of last resort, and non-judicial foreclosure, as well as in counties with more socially vulnerable populations. We further find that instrumented non-renewals are associated with reduced housing values (especially for low-value properties), retail sales, and homeownership rates, with most of these effects being larger in high-foreclosure counties. Our findings suggest preserving housing insurability given climate change is a central challenge for policymakers.

Keywords: climate change; insurance; non-renewals; foreclosures; mortgage default

JEL codes: G22; L68; Q54; R11; R31

1. Introduction

Climate change has increasingly adversely affected U.S. households through higher premiums for and reduced availability of homeowners insurance. Most existing research and policy analysis has examined effects of climate change on homeowners insurance premiums and follow-on phenomena, such as: (1) property-casualty insurers raise premiums to reflect increased expected claim losses and more expensive hedging/reinsurance; (2) these increased insurance costs reduce households' ability to cover their overall housing costs and thus to avoid mortgage default as well as to obtain new or refinance existing mortgages; and (3) the increased insurance costs are capitalized as reductions of housing values, reducing household wealth (Climate-Related Market Risk Subcommittee, 2020; Financial Stability Oversight Council, 2023; Giglio, Maggiori, Stroebe, and Weber, 2021; Keys and Mulder, 2024; Ge, Johnson, and Tzur-Ilan, 2026).

A few recent studies document that climate change has also led insurers to reduce homeowners insurance availability by not renewing existing policies, limiting new coverage, or withdrawing altogether from climate-risky areas (Kousky, Treuer, and Mach, 2024; Oh, Sen, and Tenekedjieva, 2026). While premium increases for and non-renewals of homeowners insurance have some overlapping effects, we expect that climate-change-motivated non-renewals are inherently worse for borrowers than similarly motivated increases in premiums, for two reasons. First, premium increases simply transfer expected costs of climate change from insurers to households. In contrast, non-renewals suggest insurers cannot adequately evaluate, diversify, hedge/reinsure, or price climate risks, that is, failures of climate risk-assessment systems, insurance markets, or insurance regulation. Second, mortgage contracts require borrowers to maintain hazard insurance on the mortgaged property, one of several coverages in standard homeowners insurance policies, naming the lender as the loss payee and permitting the servicer to

force-place coverage at the borrower's expense if coverage lapses (Consumer Financial Protection Bureau, 2013). Hence, non-renewals effectively force borrowers who are not willing or able to sell their homes or pay off their mortgages to take one of three undesirable actions, the first two of which may be infeasible: (1) replace the insurance coverage, very likely at a time when other insurers are reducing their underwriting exposures to the area involved; (2) accept force-placed insurance that, is both expensive and beneficial solely to the lender (e.g., does not cover personal possessions, outbuildings, or liability); or (3) default on the terms of the mortgage contract, leading to foreclosure.

We study whether non-renewals constitute an economically important channel through which adverse effects of climate change propagate first from insurers to households, and then, in part through foreclosures, from households to local economies and communities. We also investigate policy-relevant drivers of cross-sectional variation in these effects. This examination is salient for federal and increasing numbers of state and local officials and policymakers. The U.S. Senate Budget Committee (2024) concludes that homeowners insurance non-renewals are rising sharply not only in the climate-risk poster-child states Florida, Louisiana, California, and Texas, but also in Southern New England, the Carolinas, Oklahoma, the northern Rockies, New Mexico, and Hawaii. The annual homeowners insurance non-renewal rate more than doubled from 1.10 percent in 2018 to 2.37 percent in 2022 in the highest climate-risk quintile of zip codes in the U.S. (Federal Insurance Office, U.S. Department of the Treasury, 2025), suggesting that appreciable fraction of outstanding U.S. residential mortgage debt, currently about \$13.8 trillion and 64 percent of household liabilities, is collateralized by property whose insurability has already become uncertain due to climate change. Federal Reserve Chair Jerome Powell warned Congress that, if homeowners insurance becomes unavailable in an area, that area may become difficult to

mortgage (Powell, 2025). The lack of available of insurance in an area may begin as a problem for the communities involved, but non-availability in sufficiently many areas has the potential to impair both financial and social stability (Rudebusch, 2019; Kahn, Panjwani, and Santos, 2024).

We provide the first systematic national evidence about the effects of homeowners insurance non-renewals using newly available county-year homeowners insurance non-renewals data compiled by the U.S. Senate Budget Committee. These data cover the six years from 2018 to 2023 and were provided by 23 major insurers representing roughly 65 percent of the U.S. homeowners insurance market. The data measure non-renewals directly rather than requiring them to be inferred from residual-market enrollments or other information. We merge non-renewals with county-year measures of climate-disaster damages, foreclosure filings, and three types of economic outcomes: housing prices (average and distinguishing the top, mid, and low tiers), retail sales (total and distinguishing durable from non-durable goods), and homeownership rates.

Our three primary sets of empirical analyses and findings are as follows. First, we estimate the associations between county-year homeowners insurance non-renewal rates and the average climate-disaster damages in the county and in the rest of the state during the current and prior two years. Separately examining climate-disaster damages in the county versus the rest of the state helps distinguish insurers' reactions to realized climate-disaster damages in a given county from their forward-looking evaluation of their overall portfolios of climate and regulatory risks in the state. We find that non-renewals are significantly positively associated with climate-disaster damages both in that county and in the rest of the state. In terms of economic significance, moving a county-year from the lowest to the highest quintile of recent county- (rest-of-state-) level climate damage is associated with an increase in the average non-renewal rate of about 0.033 (0.035) percentage points, 6.8 (7.2) percent of one standard deviation of that rate. We further find that

these associations are concentrated in states served by financially weaker insurers, that require pre-approval of insurance rates, and that experience more uncertain climate-disaster losses, as well as in counties with more socially vulnerable populations. These cross-sectional effects are consistent with insurers adjusting the quantity of homeowners insurance coverage they supply in a county based on the strength of their balance sheets, state-level regulatory pricing frictions, the uncertainty of climate damages in the state, and the financial and other resources of the county's population.

We then estimate the effects of county-year non-renewals on foreclosures and the three types of local economic outcomes. Because both non-renewals and these outcome variables could be affected by local climate-disaster damages or other local events, rendering non-renewal rates an endogenous predictor of foreclosure rates, we use a shift-share-style instrumental variable for the non-renewal rate. The instrument is the product of a predetermined measure of each county's *share* of the homeowners insurance market—the rank of homeowners' insurance policies per owner-occupied housing unit in 2018, the beginning of our sample period—and the natural logarithm of three-year average out-of-state climate-disaster damages (the *shift*). The idea behind the share component of this instrument is that higher exposure to homeowners insurance makes insurers more sensitive to increases in climate risk. The idea behind the shift component of the instrument is that higher out-of-state climate-disaster damages increase insurers' climate risk assessments and, for multi-state insurers, place pressure on the insurers' capitalization and underwriting capacity, while being exogenous to climate damages in the county by construction.

We find that instrumented non-renewals are significantly positively associated with foreclosure rates in both the current and following years. In terms of economic significance, a one-standard-deviation increase in the non-renewal rate is associated with a 0.73 (0.69) standard deviation rise in the current-year (following-year) foreclosure rate, consistent with non-renewals

having sizeable and persistent adverse effects on that rate. We further find that these associations are concentrated in states with fewer insurers, no insurer of last resort (e.g., a FAIR plan), and non-judicial foreclosure, as well as in counties with more socially vulnerable populations. These findings are consistent with non-renewals yielding foreclosures through the hazard-insurance covenant of mortgage contracts.

These findings also illustrate that homeowners insurance non-renewals, i.e., reductions of the *quantity* of coverage provided, yield much larger increase in foreclosures than even very large premium increases on foreclosures, i.e., increases in the *price* of coverage; for example, Ge et al. (2026) find that a \$1,000 increase in premiums increases foreclosures by only 0.016 percent. The strength of the effect of non-renewals on foreclosures (and, by extension, other adverse outcomes for the households involved) enables us to examine foreclosures as a channel for the adverse effects of non-renewals on local economic variables, as discussed next.

Lastly, we find that instrumented non-renewals are associated with adverse local economic outcomes. We find that non-renewals significantly reduce local housing values, with the effects being about twice as large for the bottom tier of housing values than for the middle and top tiers. In terms of economic significance, a one-percentage-point increase in the non-renewal rate is associated with an 8.3 percent decline in average single-family home values and a 16.5 percent decline in bottom-tier home values. We find that non-renewals significantly reduce local retail sales. A one-standard-deviation increase in non-renewals is associated with a 4.2 percent decrease in total retail sales, a 5.1 percent decrease in sales of durable goods, and a 6.2 percent decrease in sales of non-durable goods. We find that non-renewals significantly reduce the homeownership rate. A one-percentage-point increase in non-renewals reduces the homeownership rate in the following year by 1.6 percentage points. Most but not all these effects are more pronounced in

high-foreclosure counties. Most notably, the effect of non-renewals on sales of durable goods is entirely through foreclosure, while the effect of sales of non-durable goods is unrelated to foreclosure.

Collectively, our evidence traces a cascade of adverse effects from climate-disaster damages to homeowners insurance non-renewals, from non-renewals to home foreclosures, and from non-renewals and foreclosures to reduced housing values, retail sales, and home ownership rates. These findings contribute to the research on the effects of climate change on insurance markets discussed above.

Our evidence also contributes to the literature on spillovers from mortgage defaults and home foreclosures to local economic variables. Existing work typically focuses on defaults arising from negative equity, credit-market disruptions, and macroeconomic shocks, and shows that the resulting foreclosures depress nearby housing prices and local demand (Campbell, Giglio, and Pathak, 2011; Mian, Sufi, and Trebbi, 2015; Guren and McQuade, 2020). We identify insurance non-renewals as a distinct and powerful trigger for foreclosure that connects climate change to foreclosures and adverse local economic consequences.

Finally, the paper informs policy debates regarding how insurance rate regulation, residual-market design, macroprudential policy, and hardening infrastructure and other assets to climate disasters can mitigate the adverse effects of climate change. Rate regulation (i.e., suppression of coverage *prices*) may protect households from sudden premium increases, but if not paired with adequate residual-market capacity, climate risk mitigation, or support for vulnerable households (i.e., maintenance of the *quantity* of coverage available), inadequate rates may lead insurers to reduce their coverage to the detriment of the affected households and their communities.

The remainder of the paper is as follows. Section 2 reviews the related literature. Section 3 describes the data and measurement. Sections 4 through 6 present results on non-renewals, foreclosures, and local economic outcomes, respectively. Section 7 reports robustness tests, and Section 8 concludes.

2. Related Literature

2.1 Climate Risk, Insurance Prices, and Insurance Availability

This paper builds on a growing climate-finance literature that studies how climate change has increased insurance premiums, made premiums more sensitive to climate risk, reduced housing values, and influenced households housing insurance, financing, and investment decisions. For example, Giglio, Maggiori, Stroebe, and Weber (2021) show that long-run climate risk reduces housing values. Keys and Mulder (2024) use mortgage escrow data to estimate homeowners insurance premiums by zip code and find these premiums are increasing and becoming more sensitive to climate-disaster risk. Ge, Johnson, and Tzur-Ilan (2026) show that higher homeowners insurance premiums increase relocation to areas with lower insurance costs by less financially constrained households and mortgage delinquency by more constrained households. Ge, Lam, and Lewis (2025) show that the removal of flood-insurance subsidies reduces housing values and mortgage financing and increases rebuilding by homeowners to increase flood resilience. Sastry, Sen, and Tenekedjieva (2024) and Sastry, Scharlemann, Sen, and Tenekedjieva (2025) provide evidence that climate change has led to households having lower quality insurance coverage due in part to financially constrained households purchasing limited coverage and in part to increasingly undercapitalized and poorly diversified insurers in climate-risky areas like Florida.

Most of the literature examining the effects of climate change on homeowners insurance focuses on the premiums charged for that insurance rather than on the availability of the insurance.

This focus assumes that offering homeowners insurance in a geographical area is not subject to frictions large enough to make it unprofitable or otherwise infeasible for insurers to offer the insurance profitably. Climate change could yield such large frictions for at least three reasons. First, at least in some climate-risky areas, climate change is a prime example of Knightian uncertainty (unknown unknowns) that makes risk-assessment difficult or impossible based on historical data and existing climate models (Kelly and Tan, 2015; Kousky, Treuer, and Mach, 2024). Second, and relatedly, such uncertainty reduces and may eliminate primary insurers' ability to manage risk through more expensive, more restrictive, or unavailable reinsurance or other means of hedging, such as catastrophe bonds (Moody's Ratings, 2023). Third, states typically regulate homeowners insurance premiums either through required prior approval of premiums or ex post evaluation of whether premiums are excessive or unfairly discriminatory, and these regulatory decisions typically are based to a significant extent on insurers' historical claim losses (GAO, Appendix IV, 2026).

When insurers deem such frictions too large, they may not renew policies, reduce new underwriting, or withdraw altogether from a geographical area (Kousky et al., 2024; Oh et al., 2026). Our study therefore complements studies that focus on increase in the *price* of homeowners insurance coverage by examining how insurers manage their exposure to climate change by reducing the *quantity* of coverage they supply by not renewing policies, as well as the effects of such non-renewals.

2.2 Hazard-Insurance Covenants, Technical Defaults, and Foreclosures

The hazard-insurance covenant in standard mortgage contracts is the primary reason why non-renewals of homeowners insurance policies lead to foreclosures and other adverse consequences for the households involved. Hazard insurance is one of several coverages in

homeowners insurance policies; hazard insurance covers only the mortgaged property, not personal possessions, outbuildings, or liability. For a mortgaged property, hazard insurance names the mortgage lender as the loss payee. When a mortgage borrower fails to maintain the required insurance—say because the homeowners insurance policy is not renewed and replacement insurance is either unavailable or more expensive than the household can manage—technical default on the mortgage occurs. Subject to federal rules that require mortgage servicers to have a reasonable basis for believing that the borrower has not obtained hazard insurance (Consumer Financial Protection Bureau, 2013), mortgage servicers may obtain force-placed insurance, which typically is both costly for the borrower and protects only the value of the lender’s lien on the mortgaged property rather than satisfying the borrower’s overall insurance needs (Cronkite, 2016). While foreclosure is a costly process, lenders may instead foreclose on the mortgaged property if they deem the borrower likely to default or otherwise view foreclosure as preferable to maintaining the mortgage.

Technical default on the hazard-insurance covenant due to non-renewal of a homeowners insurance policy constitutes a pathway to mortgage non-compliance that is distinct from payment default. Standard payment-default models in the mortgage setting emphasize a double trigger: borrowers default when they have negative equity in the mortgaged property and experience an adverse financial shock (Gerardi, Herkenhoff, Ohanian, and Willen, 2018). In contrast, non-renewal-based technical default can occur without either negative home equity or a financial shock other than the higher cost or reduced availability of homeowners insurance.

2.3 Adverse Effects of Foreclosure on Local Economic Variables

A separate related literature studies the adverse effects of foreclosures on local property values and other local economic variables. Campbell, Giglio, and Pathak (2011) show that

foreclosed properties sell at substantial discounts and depress nearby home values. Anenberg and Kung (2014) and Gerardi, Rosenblatt, Willen, and Yao (2015) provide evidence that the effects of foreclosures on local property value are due lack of property maintenance rather than demand or supply effects. In contrast, Guren and McQuade (2020) find that the large volume of foreclosures in the wake of the financial crisis lead to a downward spiral in house prices due to the reduced equity of lenders, the ruined credit of the borrowers involved, and choosier buyers. Mian, Sufi, and Trebbi (2015) show that foreclosures are higher in non-judicial foreclosure states and reduced house prices, residential investment, and household consumption during the financial crisis.

2.4 Hypotheses

Our hypotheses are motivated in part by the literature examining the effects of climate change on homeowners insurance premiums and availability discussed in Section 2.1 and the literature examining the effects of foreclosures on local economic variables discussed in Section 2.3, as well as by the hazard insurance covenant in mortgage contracts discussed in Section 2.2. As discussed in the introduction, these hypotheses pertain to a cascade of three sets of adverse effects: (1) from climate-disaster damages to homeowners insurance non-renewals, (2) from non-renewals to home foreclosures, and (3) from non-renewals and foreclosures to reduced housing values, retail sales, and home ownership rates.

We expect that climate-disaster damages lead insurers to update their assessments of the risk and future profitability of providing coverage in the affected areas and in areas with similar climate risk, and that sufficiently large assessed increases in risk and decreases in future profitability lead insurers to reduce their exposures by not renewing policies in these areas. Because insurers are capitalized and regulated and homeowners insurance is regulated at the state level, and because the counties in states often exhibit correlated climate risks, we expect insurers

make their coverage decisions jointly across the counties in a state. Thus, we expect that climate-disaster damages in the county and the rest of the state increase homeowner insurance non-renewals in the county. We further expect that these increases are concentrated in states served by financially weaker insurers, that require pre-approval of insurance rates, and that experience more uncertain climate-disaster losses, as well as in counties with more socially vulnerable populations.

Due to the hazard insurance covenant in mortgage contracts, we expect homeowners insurance non-renewals sometimes lead to increased foreclosures on the mortgaged properties. We further expect these increases in foreclosures are larger when there are fewer alternative insurance providers in the state, no residual-market backstop is available in the state, and state foreclosure law allows lenders to begin the foreclosure process sooner, as well as when the county population is more socially vulnerable.

In part through their effects on foreclosure, we expect homeowners insurance non-renewals in a county to reduce housing values and thus household wealth and the collateral for mortgages, investments in maintaining existing or building new housing, retail sales, and homeownership in that county. Moreover, because financially constrained households are more likely to occupy lower-value homes and to reduce non-essential consumption when they experience adverse financial shocks, we further expect the effects of non-renewals on housing values to be concentrated in the lower part of the housing distribution. Because foreclosures are among the most adverse effects of non-renewals, we further expect that these effects are larger in high-foreclosure counties.

3. Data Sources, Variables, Sample, and Descriptive Statistics

We conduct the empirical analyses at the county-year level. We obtain and merge county-year level data on homeowners insurance non-renewals, climate-disaster damages, foreclosures, housing prices, retail sales, homeownership, and economic and demographic characteristics.

3.1 Data Sources and Variables

Our primary insurance-market data come from the U.S. Senate Budget Committee’s homeowners insurance non-renewal dataset. In response to concerns about instability in the U.S. homeowners insurance market, the Committee requested information from 41 major property-casualty insurers that underwrite homeowners insurance coverage. Twenty-three insurers, representing approximately 65% of the U.S. homeowners insurance market, provided the requested data. The dataset includes county-year data on homeowners insurance policies and non-renewals each year from 2018 to 2023. To our knowledge, it is the first public national dataset that measures homeowners insurance non-renewals at a geographically disaggregated level, making it possible to conduct studies of non-renewals in almost all local insurance markets.

We specify $NonrenewalRate_{c,t}$ as the number of non-renewed homeowners insurance policies divided by the number of policies in force prior to these non-renewals, expressed in percentage points, in county c in year t . This measure captures insurer-initiated withdrawals from existing policies rather than reduced underwriting of new policies or complete withdrawal from the county. Because very small numbers of policies in force likely yield noisy non-renewal rates, we exclude county-years with fewer than 500 policies in force.

We obtain the dollar amounts of climate-disaster damages to property from the Spatial Hazard Events and Losses Database for the United States (SHELDUS). We exclude crop damages as well as damages from earthquakes, tsunamis, volcanos, and other non-climate disasters. We

also exclude floods because flood insurance coverage is excluded from standard homeowners insurance policies and typically provided through the National Flood Insurance Program.¹ Thus flood risk is not directly tied to insurers' homeowners insurance coverage and pricing decisions. Because the dollar amounts of climate-disaster damages are larger in counties with more structures, higher property values, and greater economic activity, all else equal, we scale these dollar amounts by real GDP obtained from the Bureau of Economic Analysis, which yields measures of damage intensity (Nordhaus, 2010; Oh, Sen, and Tenekedjieva, 2026).

We average the climate-disaster damage variables across the three years ending with the current year. *LocalDamage_raw_{c,t}* is the average of annual climate-related property damage scaled by real GDP in county *c* across years *t-2* to *t*. *StateOtherDamage_raw_{c,t}* is the average of annual climate-related property damage in all other counties in county *c*'s state, excluding county *c*, scaled by the real GDP of these counties across year *t-2* to *t*. These raw climate-disaster damage measures are highly skewed right. To mitigate the potential for our results to be driven by outliers and because insurers' responses to climate-disaster damages may exhibit nonlinearities (e.g., insurers may tolerate small losses but respond more strongly to sufficiently severe losses), in the empirical models we use the within-year quintile ranks of the two raw climate-disaster damage variables, scaled from zero to one, denoted *LocalDamage_{c,t}* and *StateOtherDamage_{c,t}*.

We obtain annual foreclosures by zip code, which we aggregate to the county-year level, from ATTOM, formerly RealtyTrac, which collects foreclosure-related filings from public records. ATTOM records multiple stages of the foreclosure process, including early-stage default notices and later-stage sale notices. Depending on whether a state requires judicial foreclosure, lenders submit one of the following two types of filings appreciably in advance of any foreclosure auction:

¹ Mortgage contracts generally require flood insurance mortgaged properties in Special Flood Hazard Areas, and these contracts may, at the lender's discretion, require flood insurance in other flood-prone areas.

a Notice of Default in non-judicial foreclosure states and a Lis Pendens in judicial foreclosure states. After these filings are made, borrowers may cure their defaults, say by paying off the mortgages by selling the mortgaged properties or out of household assets, in which case foreclosure auctions do not occur. If borrowers do not cure their defaults, lenders submit one of the following two types of filings shortly before any foreclosure auction: a Notice of Trustee Sale (NTS) in non-judicial foreclosure states and a Notice of Foreclosure Sale (NFS) in judicial foreclosure states. We follow the approach in the foreclosure literature (Mian et al., 2015) and measure foreclosures using NTS and NFS filings, as these filings occur close to the foreclosure auction and therefore more directly capture realized foreclosure events than earlier notices, which may be cured before an auction occurs. We define *Foreclosure* as the sum of these sale-related filings divided by the number of residential housing units with mortgages in the county, expressed in percentage points.

We measure housing values using Zillow's Home Value Indices (ZHVI). For each county-year, we calculate the natural logarithm of the annual average ZHVI for single-family homes, as well as the natural logarithms of Zillow's ZHVI measures for the bottom (5th to 35th percentile), middle (35th to 65th percentile), and top (65th to 95th percentile) tiers of the housing-value distribution. These tiered measures allow us to examine whether insurance-market stress has stronger effects on lower-value homes, which are more likely to have financially constrained owners who are less able to maintain mortgage compliance and thus subject to greater pressure to sell their properties.

We measure local consumption using establishment-level retail sales from Data Axle. The main measure is the natural logarithm of total county-year retail establishment sales. We exclude wholesale sales and focus on establishments with primary four-digit SIC codes in the retail sector.

We also separate total sales into sales of durable and non-durable goods. Durable goods include building materials and hardware, motor vehicles, furniture and other home furnishings, and shopping goods such as jewelry. Non-durable retail sales include groceries, apparel, gasoline and other fuels, and the items sold at liquor, drug, and variety stores and eating and drinking establishments.

We obtain homeownership data from the U.S. Census Bureau's American Community Survey. We measure the county-year homeownership rate as owner-occupied housing units divided by total occupied housing units, expressed in percentage points. We also examine the natural logarithms of owner-occupied units and total occupied units. This decomposition helps facilitate interpretations because decreases in the homeownership rate can result from homeowners either becoming renters in the county or leaving the county; homeowners becoming renters in the county does not reduce the total occupancy in the county, while homeowners leaving the county does.

We supplement the main variables described above with controls for county demographic and economic characteristic as well as the variables used in cross-sectional tests. The control variables are measured in the prior year, and include the shares of residents who are Black or aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. These variables come from the Census, the Bureau of Labor Statistics, and the Bureau of Economic Analysis.

The variables used in the cross-sectional tests are as follows. We obtain insurers' financial characteristics and homeowners insurance market shares in each state-year from the NAIC. We measure insurance pricing frictions using state homeowners insurance rate-filing rules provided by the NAIC, defining high-friction states as those that subject homeowners insurance rates to

prior approval.² We measure county-level social vulnerability (i.e., lack of resilience to stressors of various sorts) using the CDC/ATSDR Social Vulnerability Index, denoted *SVI*, which combines information on socioeconomic status, household characteristics, housing and transportation, and racial and ethnic minority status. *SVI* is used by policymakers, public health officials, and emergency managers to allocate resources and prepare or respond to disasters, and it has been widely used in research on climate and other disasters (e.g., Ramesh et al., 2022; Billings, Ryan, and Yan, 2026). All continuous variables used in the regression analyses are winsorized at the 1st and 99th percentiles. Appendix A provides the definitions and data sources for all variables.

3.2 Sample and Descriptive Statistics

We construct the sample starting with county-year data on homeowners insurance non-renewals from the U.S. Senate Budget Committee. We exclude the District of Columbia and county-years with fewer than 500 homeowners insurance policies in force, which yields non-renewals for 16,396 county-years for 2,768 distinct counties from 2018 to 2023.

We then merge non-renewals with county-year data on climate-disaster damages, foreclosures, local economic outcomes, and control variables. The analysis of climate-disaster damages and non-renewals is based on a sample of 16,281 county-years for 2,743 counties; 115 county-year observations and 25 counties are lost due to missing values in lagged county characteristic controls. For the analysis of non-renewals and foreclosures, the construction of the share-shift-style instrumental variable excludes 50 counties without data on non-renewals in 2018, yielding a sample of 16,109 county-year observations (a loss of 172) for 2,693 counties. For the analyses of non-renewals and local economic variables, sample sizes vary with the availability of

² Fourteen states require prior approval for (average) homeowners insurance rates; some of these states may allow exceptions for small rate changes or similar reasons.

the outcome variable. The sample for the Zillow housing-value analysis contains 15,759 county-years for 2,648 counties, for the Data Axle retail-sales analysis contains 16,096 county-years for 2,691 counties, and for the ACS homeownership analysis contains 16,109 county-years for 2,693 counties.

Figure 1 depicts cross-sectional and temporal variation in the homeowners insurance non-renewal rate. Panel A depicts the non-renewal rate by county in 2023, the most recent sample year. Florida, particularly the Gulf Coast and most of the peninsula, represents the largest concentration of high non-renewal rates in the country. Sizeable concentrations of high rates are also visible along the Gulf Coast in Louisiana and Texas and in the coastal Carolinas, the mountainous northern and eastern portions of California, and Oklahoma. Smaller concentrations appear in scattered counties throughout most of the West and western Midwest. In contrast, most of the eastern portion of the Midwest and Northeast outside of a few small coastal areas exhibit low non-renewal rates.

Panel B depicts the changes in county-level non-renewal rates between 2018 and 2023. The largest concentrations of increases in non-renewal rates generally are in the same areas with concentrations of high non-renewal rates (e.g., Florida, the coastal Carolinas, the mountainous northern and eastern parts of California, and Oklahoma), while the eastern Midwest and Northeast counties, areas with low non-renewal rates, generally exhibit modest changes in non-renewal rates. Hence, increases in non-renewal rates overlap substantially with the levels of the rates, consistent with insurers primarily reducing their underwriting of homeowners insurance in climate-risky areas. Our county-year research design exploits both cross-sectional and time-series variation in non-renewal rates.

Table 1 reports descriptive statistics for the main model variables. The average homeowners insurance non-renewal rate is 0.783 percentage points, with a median of 0.680

percentage points and a standard deviation of 0.484 percentage points. Thus, non-renewals are not rare events and exhibit meaningful variation across the sample county-years. The average foreclosure rate is 0.475 percentage points in year t and 0.399 percentage points in year $t+1$. Hence, non-renewals are more common than foreclosures, presumably in part because non-renewals often lead to households obtaining insurance from another provider, sales of the property involved, or mortgage prepayments out of other household assets.

The raw climate-damage measures are highly skewed right; the mean of *LocalDamage_raw* of 0.089 is thirty times its median of 0.003, while the mean of *StateOtherDamage_raw* of 0.079 is seven times its median of 0.011. This skewness reflects the concentration of climate-disaster damages in relatively few high-damage disasters. The skewness motivates our use of within-year quintile rank variables, *LocalDamage* and *StateOtherDamage*, in the empirical analysis.

The average natural logarithm of single-family home value is 12.208 (\$200,400) in the overall sample and ranges from 11.675 (\$117,600) in Zillow's bottom tier to 12.732 (\$338,400) in its top tier. The mean homeownership rate is 72.398 percentage points. All the local economic variables exhibit substantial variation that helps to enhance the power of our tests of the effects of non-renewals on these variables.

The mean unemployment rate is 4.720%, reflecting the generally strong labor-market conditions over much of the sample period, apart from the massive but short spike at the onset of the COVID-19 pandemic. The distribution of the Black population share is right-skewed, with a mean of 10.5% and a median of 3.7%. The average share of residents aged 20 to 44 is 30.0%, and the average natural logarithm of income per capita is 10.740 (\$46,200). All the economic and

demographic control variables exhibit substantial variation, consistent with meaningful heterogeneity across counties.

Internet Appendix Table IA1 reports the annual means and medians of non-renewal and foreclosure rates. Both series exhibit U-shaped patterns from 2018 to 2023, with low values during the COVID-19 pandemic and higher values before and afterwards. The troughs in the pandemic are consistent with pandemic-era moratoria on adverse actions against households.³ The mean and median non-renewal rates more than double from 2020 to 2023, consistent with tightening of homeowners insurance markets as pandemic restrictions faded and climate risks intensified.

4. Effects of Climate-Disaster Damages on Homeowners Insurance Non-Renewals

4.1. Baseline Estimations

We first test the hypotheses stated in Section 2.4 regarding whether recent climate-disaster damages lead insurers to increase their homeowners insurance non-renewal rates. To test these expectations, we estimate the following county-year specification:

$$\begin{aligned} NonrenewalRate_{c,t} = & \beta_1 LocalDamage_{c,t} + \beta_2 StateOtherDamage_{c,t} \\ & + \Gamma X_{c,t-1} + \alpha_c + \delta_t + \varepsilon_{c,t}. \end{aligned} \quad (1)$$

The dependent variable, $NonrenewalRate_{c,t}$, and the explanatory variables of interest, $LocalDamage_{c,t}$ and $StateOtherDamage_{c,t}$, are defined in Section 3.1. The coefficient β_1 captures insurers' responses to climate-disaster damages in the same county as insurers provide homeowners insurance coverage. The coefficient β_2 captures insurers' responses to climate-disaster damages in the rest of the state. We expect both coefficients to be positive.

³ ATTOM (2021) quotes Rick Sharga, the executive vice president of RealtyTrac, as explaining the low rate of foreclosure during the COVID-19 pandemic: "The [federal] government's moratoria have essentially stopped foreclosure activity on everything but vacant and abandoned properties." Some states made it difficult to cancel homeowners and other insurance policies during the pandemic; see, for example, New York State Department of Financial Services (2020).

The vector of control variables $X_{c,t-1}$ includes one-year lagged demographic and economic characteristics for county c : the share of Black residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. County fixed effects α_c absorb all time-invariant county-specific factors and year fixed effects δ_t absorb all county-invariant year-specific factors in the sample.

Table 2 presents three OLS estimations of equation (1); in columns (1) and (2), $LocalDamage_{c,t}$ and $StateOtherDamage_{c,t}$ are included separately, while in column (3) these variables are both included. Both climate-damage variables are at least weakly significantly positively associated with the non-renewal rate in all columns. For example, in column (3), the coefficient on $LocalDamage$ is 0.033 ($p < 0.1$). This coefficient implies that the highest quintile of recent local climate damage has a 0.033 percentage-point higher homeowners insurance non-renewal rate than the lowest quintile, all else equal. This difference is about 4 percent of the mean non-renewal rate and about 7 percent of one standard deviation of that rate. Thus, as expected, insurers increase their homeowners insurance non-renewal rates in counties that experience climate-disaster damages.

The coefficient on $StateOtherDamage$ in column (3) is very similar at 0.035 ($p < 0.05$). Thus, insurers increase their non-renewal rates in a county when other counties in the same state experience climate-disaster damages. The result is consistent with our expectation that insurers make their homeowners insurance coverage decisions at the state level. In summary, the results reported in Table 2 show that recent climate-disaster damages in a county and the remainder of the state increase the homeowners insurance non-renewal rate in the county.

4.2 Cross-sectional Tests

We conduct cross-sectional tests to examine four channels that we expect explain the baseline results reported in Table 2: state-level insurer capitalization, regulatory pricing frictions, and uncertainty about climate risk, as well as county-level social vulnerability. The pairwise Pearson correlations of our proxies for these constructs, described below, range from 0.046 to 0.173, indicating the constructs are largely distinct.

Regarding the state-level insurer capitalization channel, climate-disaster damages in the county and remainder of the state generate claim losses, thereby reducing insurers' capital. To the extent that homeowners insurance non-renewals reflect limited insurer capital, we expect the effect of climate-disaster damages on non-renewals is stronger in states served by financially weaker insurers. We measure insurer financial weakness using a state-level indicator, *WeakCapital*, that equals one for states with below-median weighted-average insurer risk-based capital ratios and zero otherwise.

Column (1) of Table 3 reports the OLS estimation of an expansion of equation (1) that includes *WeakCapital* separately and interacted with both *LocalDamage_{c,t}* and *StateOtherDamage_{c,t}*. As expected, the coefficients on the interactions of *WeakCapital* with both *LocalDamage_{c,t}* (0.043, $p < 0.05$) and *StateOtherDamage_{c,t}* (0.094, $p < 0.01$) are both significantly positive, with the latter coefficient being over twice as large, consistent with insurers' capitalization being determined at the state level. Thus, relatively weakly capitalized insurers increase their homeowners insurance non-renewal rates more after experiencing climate-disaster damages in the county and the rest of the state. Moreover, the coefficients on the uninteracted *LocalDamage_{c,t}* and *StateOtherDamage_{c,t}* are both insignificant and close to zero, consistent

with relatively well capitalized insurers not increasing non-renewal rates after experiencing climate-disaster damages.

Regarding the state-level regulatory pricing-friction channel, we expect that insurers usually prefer to respond to climate-disaster damages by increasing premiums rather than by reducing coverage, but they may be constrained from increasing premiums by state regulation. When this is the case, we expect insurers to reduce coverage instead, in part by not renewing homeowners insurance policies when they expire. We measure regulatory pricing frictions using a state-level indicator, *Frictions*, that takes a value of one for states in which homeowners insurance rate increases are subject to prior approval by state regulators and zero otherwise. *Frictions* does not vary over time.

Column (2) of Table 3 reports the estimation of an expansion of equation (1) that includes the interactions of *Frictions* with both *LocalDamage_{c,t}* and *StateOtherDamage_{c,t}*. As expected, the coefficient on the interaction of *Frictions* and *StateOtherDamage_{c,t}* (0.137, $p < 0.01$) is positive and statistically significant, consistent with insurers renewing policies at a lower rate when they cannot raise premiums enough to reflect increases in assessed climate risk. However, the coefficient on the interaction of *Frictions* and *LocalDamage_{c,t}* is insignificant; this may reflect state-level approvals regarding premium increases being more sensitive to state-level factors.

Regarding state-level uncertainty about climate risk, we use the within-year quintile ranks of the coefficient of variation of climate-disaster damages scaled by GDP for the state over the prior three years, rescaled to range from 0 to 1, denoted *Uncertainty*. We use *Uncertainty* as a proxy for unquantifiable Knightian uncertainty (unknown unknowns), as opposed to quantifiable Knightian risk (known unknowns). Consistent with theoretical findings (e.g., see Easley and

O'Hara 2009) that Knightian uncertainty is more difficult to price than Knightian risk, and thus yields either an additional increment to required returns or an inability of parties to agree on the terms of trade, as well as recent research on the implications of insurers' limited ability to evaluate climate risk (Boomhower, Fowle, Gellman, and Plantinga, 2025), we expect insurers to be more likely to not renew homeowners insurance policies when climate-disaster damages occur in states with more uncertainty about future damages.

Column (3) of Table 3 reports the estimation of an expansion of equation (1) that includes *Uncertainty* separately and interacted with both *LocalDamage_{c,t}* and *StateOtherDamage_{c,t}*. As expected, the coefficients on the interactions of *Uncertainty* with both *LocalDamage_{c,t}* (0.097, $p < 0.01$) and *StateOtherDamage_{c,t}* (0.477, $p < 0.01$) are significantly positive, with the latter coefficient being over almost five times as large. Thus, insurers subject to more uncertainty increase their homeowners insurance non-renewal rates more after experiencing climate-disaster damages. While it may seem surprising that the coefficient on *Uncertainty* is significantly negative (-0.203, $p < 0.01$), this coefficient is significantly positive (0.071, $p < 0.01$) if the interaction terms in which it appears are omitted, so the sign of this coefficient appears to be attributable to non-linearity in the relationship. Non-linearity also appears to be the reason why the coefficient on *StateOtherDamage_{c,t}* is significantly negative (-0.296, $p < 0.01$).

Regarding the county-level social vulnerability channel, we use the county-level Social Vulnerability Index, *SVI*, described in Section 3.1 to capture counties' resilience to adverse shocks, including climate disasters. We expect insurers to be more sensitive to assessed increases in climate risk in socially vulnerable counties because these counties tend to have infrastructures and insured properties are less resilient to climate disasters, in part due to a history of "redlining" (discriminatory lending) in many of those areas (Lu et al., 2025).

Column (4) of Table 3 reports the estimation of an expansion of equation (1) that includes *SVI* separately and interacted with both *LocalDamage_{c,t}* and *StateOtherDamage_{c,t}*. As expected, the interactions of *SVI* with both *LocalDamage_{c,t}* (0.125, $p < 0.05$) and *StateOtherDamage_{c,t}* (0.373, $p < 0.01$) are significantly positive. The coefficient on *StateOtherDamage_{c,t}* is over three times larger than the coefficient on *LocalDamage_{c,t}*, consistent with insurers being disproportionately likely to not renew policies in socially vulnerable areas when climate disasters occur elsewhere, a behavior suggestive of “bluelining” (i.e., withdrawal of insurance coverage and other financial services from areas with low resilience to climate risk, often because these areas were previously subject to redlining) (Salazar-Miranda, Conzelmann, and Phan, 2024). Providing support for this suggestion, the coefficient on the uninteracted *StateOtherDamage_{c,t}* (-0.178, $p < 0.01$) is significantly *negative*, consistent with socially advantaged counties experiencing *lower* non-renewal rates when climate disasters occur elsewhere in the state.

In summary, the results reported in Table 3 indicate that the effects of climate-disaster damages on homeowners insurance non-renewals are amplified when the insurers serving a state are less well capitalized, state-level regulatory constraints on homeowners insurance premiums are higher, and county-level social vulnerability is higher. Thus, insurers’ decisions to not renew homeowners insurance policies are more than mechanical responses to climate-disaster damages.

5. Effects of Homeowners Insurance Non-Renewals on Foreclosures

We next test the hypotheses stated in Section 2.4 that increases in homeowners insurance non-renewal rates in a county lead to increased foreclosures in the county due to the hazard insurance covenant in mortgage contracts. As preliminary descriptive evidence of this association, Figure 2 plots four binned scatterplots of the level and change in foreclosure rates in years t and

$t+1$ against non-renewal rates in year t . The non-renewal rate and foreclosure rate variables are positively associated in all four plots. While these plots do not address the likelihood that both rates could be affected by local climate-disaster damages or other local events, rendering non-renewal rates an endogenous predictor of foreclosure rates, they show that the association is robust to both the choice of horizon and first differencing. As discussed below, to address concerns about the endogeneity of the non-renewal rate directly, we test these expectations by conducting regression analysis and instrumenting for the non-renewal rate using a shift-share-style instrumental variable.

5.1. Identification Strategy

We estimate the following county-year regression specification:

$$Foreclosure_{c,t+k} = \beta NonrenewalRate_{c,t} + \Gamma X_{c,t-1} + \alpha_c + \delta_t + \varepsilon_{c,t+k}, \quad k \in \{0,1\}. \quad (2)$$

The dependent variable $Foreclosure_{c,t+k}$ is the foreclosure rate in county c in year t ($k = 0$) or year $t + 1$ ($k = 1$). We examine both contemporaneous and one-year-ahead foreclosure rates because the consequences of losing homeowners insurance coverage on foreclosure may be subject to delays for various reasons, such as the regulatory requirement for mortgage servicers to have a reasonable basis for believing that the borrower has not obtained hazard insurance.

The instrument for the explanatory variable of interest, $NonrenewalRate_{c,t}$, equals the product of a predetermined measure of each county's *share* of the national homeowners insurance market—the percentile rank of homeowners' insurance policies per owner-occupied housing unit in 2018, the beginning of our sample period, denoted $Exposure_{c,2018}$ —and the natural logarithm of three-year average out-of-state climate-disaster damages, denoted $OutStateDamage_{c,t}$ (the shift).

$$IV_{c,t} = Exposure_{c,2018} \times OutStateDamage_{c,t}. \quad (3)$$

The idea behind the share component of this instrument is that higher exposure to homeowners insurance makes insurers more sensitive to increases in climate risk. The idea behind the shift component of the instrument is that higher out-of-state climate-disaster damages increase insurers' climate risk assessments and, for multi-state insurers, place pressure on the insurers' capitalization and underwriting capacity, while being exogenous to climate damages in the county by construction. Because the share component of the instrument is based on 2018 data, the IV sample excludes the approximately 1.1 percent of counties for which the non-renewal rate is missing in that year. These counties are mostly geographically small or rural.

Equation (2) includes the same fixed effects and control variables as in equation (1).

5.2. Baseline Estimations

Table 4 reports the baseline two-stage least-squares (2SLS) estimations. The first stage estimation indicates the instrument is strong: the Kleibergen-Paap rk Wald F statistic of 58.44 lies well above the Stock and Yogo (2005) 10 percent maximal-IV-size critical value of 16.38. Internet Appendix Table IA2 documents a strongly positive first-stage coefficient (0.42, $t > 7$) that is robust to controls. The second-stage coefficient on the instrumented non-renewal rate is 0.917 ($p < 0.01$) for the contemporaneous foreclosure rate and 0.711 ($p < 0.01$) for the one-year-ahead rate. Internet Appendix Figure IA1 reports the coefficient by month during year $t+1$ and shows that the coefficient is positive at every monthly horizon.

In terms of economic significance, a one-standard-deviation increase in the non-renewal rate (0.484 percentage points) raises the contemporaneous foreclosure rate by approximately 0.44 ($= 0.484 \times 0.917$) percentage points, roughly three-quarters of the standard deviation of the foreclosure rate. The contemporaneous coefficient exceeds the one-year-ahead coefficient, which

makes sense because covenant violations are most readily acted upon while the violations are fresh, and the borrowers most affected by a loss of coverage are often already financially stressed.

These results suggest that increased non-renewals of homeowners insurance yield increased home foreclosures due to the hazard insurance covenant in mortgage contracts. To our knowledge, these are the first systematic causal estimates that identify non-renewals as a material driver of foreclosures, separate from the direct effect of climate-disaster damages on borrowers' ability to make mortgage payments.

5.3 Cross-Sectional Tests

The baseline estimation capture the average effect of homeowners insurance non-renewals on foreclosures. We next conduct cross-sectional tests to identify circumstances where this effect is stronger. We expect that non-renewals have larger effects on foreclosures when fewer insurers operate in the state from which to obtain replacement insurance (measured using both the number of insurers and the Herfindahl market concentration index), when there is no insurer of last resort in the state,⁴ when the state allows non-judicial foreclosure, and in more socially vulnerable counties (i.e., when *SVI* is higher) in which households have lower resilience to adverse events.

Table 5 reports estimations of expansions of equation (2) that include these cross-sectional variables both separately and interacted with *NonrenewalRate*. To avoid the forbidden regression problem in interactive 2SLS models, we instrument for the interaction of *NonrenewalRate* with each partitioning variable using the shift-share-style instrument times that partitioning variable. Panel A (B) reports the estimations with current-year (one-year-ahead) foreclosure rates as the dependent variable. The two panels provide the following consistent

⁴ For example, see the website of the California Fair Plan (<https://www.cfpnet.com/key-statistics-data/>) for statistics on the increasing reliance of California homeowners on the plan as an insurer of last resort.

evidence. First, in column (1) [column (2)] of both panels, the coefficient on instrumented interaction of *NonrenewalRate* with *#Insurers* [*HHI*] is significantly negative [positive], consistent with non-renewals have stronger effects on foreclosures when fewer insurers operate in the state [the homeowners insurance market is more concentrated]. Second, in column (3) of both panels, the coefficient on the instrumented interaction of *NonrenewalRate* with *NoFairPlan* is significantly positive, consistent with the presence of insurers of last resort in the state reducing the probability that non-renewals lead to foreclosures. Third, in column (4) of both panels, the coefficient on the instrumented interaction of *NonrenewalRate* with *Nonjudicial* is significantly positive, consistent with hazard insurance covenant breaches leading more quickly to foreclosure filings in states with non-judicial foreclosure. Lastly, in column (5) of both panels, the coefficient on the instrumented interaction of *NonrenewalRate* with *SVI* is significantly positive, consistent with non-renewals have stronger effects on foreclosures in more socially vulnerable counties.

In summary, the results of the cross-sectional tests reported in Table 5 strengthen the causal interpretation of the baseline estimations. Non-renewals of homeowners insurance policies increase foreclosures most strongly under the expected conditions: fewer alternative providers of insurance, the absence of an insurer of last resort, more rapid legal processes for foreclosure, and more socially vulnerable households.

6. Effects of Homeowners Insurance Non-Renewals on Local Economic Variables

We next test the hypotheses stated in Section 2.4 that increases in homeowners insurance non-renewal rates in a county lead to increased homeowners insurance non-renewals in a county lead to decreases in housing values, retail sales, and homeownership. To test these hypotheses, we estimate the following county-year regression specification:

$$Y_{c,t+1} = \beta NonrenewalRate_{c,t} + \Gamma X_{c,t-1} + \alpha_c + \delta_t + \varepsilon_{c,t+1}, \quad (4)$$

where $Y_{c,t+1}$ is a real local economic outcome in county c in year $t+1$. We instrument for $NonrenewalRate_{c,t}$ using the shift-share-style instrument specified in equation (3). Equation (4) includes the same fixed effects and control variables as in equations (1) and (2).

6.1. Housing Prices

Table 6 reports the estimations of equation (4) with $Y_{c,t+1}$ specified as the natural logarithm of ZHVI for single-family homes in column (1) and for the bottom, middle, and top tiers of the county housing-price distribution in columns (2) to (4), respectively. We expect the depressing effect of homeowners insurance non-renewals on housing values to be strongest for the homeowners in the bottom tier. For these homeowners, it is likely that insurance from alternative providers will be more difficult to obtain and more expensive and difficult to manage when obtained, and that paying off the mortgage to eliminate the hazard insurance covenant will be less feasible. Hence, non-renewals are more likely push these homeowners into either forced sales or technical or payment defaults, raising the supply and reducing the value of bottom-tier housing.

As expected, the coefficient on the instrumented non-renewal rate is significantly negative in all four columns. Moreover, the coefficient in column (2) for the bottom tier is almost twice as negative as the coefficients in the other three columns. A one-percentage-point increase in the non-renewal rate is associated with an 8.3 percent decline in average single-family prices but a 16.5 percent decline at the bottom tier. In terms of economic significance, a one-standard-deviation increase in the non-renewal rate yields to decline in housing values of approximately 4.0 percent on average and 8.0 percent in the bottom tier. These percentages likely constitute significant shock

to the wealth of the affected households, given that home equity is the dominant asset for the typical U.S. household.⁵

6.2. Retail Sales

Table 7 reports the estimations of equation (4) with $Y_{c,t+1}$ specified as the natural logarithm of three measures of sales in county retail establishments: total sales in column (1) and sales of durable and non-durable goods in columns (2) and (3), respectively. We expect higher homeowners insurance non-renewals to reduce all three measures of sales. We expect these effects to differ in magnitude across the three measures depending upon the consequences of non-renewals for home ownership. If homeowners are able to remain in their homes by obtaining alternative insurance or paying off their mortgages, we expect the percentage decreases in sales of durable goods and non-durable goods to be similar. In contrast, if homeowners must leave their homes through foreclosure, we expect the decreases in sales to be disproportionately larger for durable goods, because a substantial portion of these goods are items most likely to be purchased by homeowners, such as building materials, appliances, furniture, and other home furnishings.

As expected, the coefficient on the instrumented non-renewal rate is at least weakly significantly negative in all three columns. In column (1) for total retail sales, the coefficient is -0.086 ($p < 0.05$). The coefficients in columns (2) and (3) for durable and non-durable goods sales are similar at -0.106 ($p < 0.10$) and -0.129 ($p < 0.01$), respectively, consistent with homeowners experiencing non-renewals often being able to stay in their homes. In terms of economic significance, a one-standard-deviation increase in the non-renewal rate is associated

⁵ For example, Kochhar and Moslimani (2023) report that in “2021, the median net worth of U.S. households overall stood at \$166,900, counting all assets. But their median net worth *without* home equity included was only \$57,900.”

with decreases of 4.2 percent in total retail sales, 5.1 percent in durable-goods sales, and 6.2 percent in non-durable goods sales.

6.3. Homeownership

Table 8 reports the estimations of equation (4) with $Y_{c,t+1}$ specified as the local homeownership rate (i.e., owner-occupied housing units divided by total occupied housing units) in column (1), as well as the natural logarithms of the numerator and denominator of the rate in columns (2) and (3), respectively. We expect higher homeowners insurance non-renewals to reduce both the homeownership rate and the natural logarithm of owner-occupied housing units. Whether non-renewals have no effect on or reduce the natural logarithm of total occupied housing units depends upon whether non-renewals lead to population loss or homelessness in the county, but at a minimum we expect that non-renewals lead to a smaller percentage decline in total occupied housing units than in owner-occupied housing units.

As expected, the coefficient on the instrumented non-renewal rate is at least significantly negative in columns (1) and (2), and it is less negative in column (3) than in column (2). The coefficient in column (1) (1.591, $p < 0.05$) indicates that a one-percent increase in the non-renewal rate leads to a 1.591 percentage-point decrease in the next-year homeownership rate. The coefficients in column (2) (-0.138, $p < 0.01$) and column (3) (-0.112, $p < 0.01$) indicate that a one-standard-deviation increase in the non-renewal rate leads to a 6.3 percent decrease in owner-occupied housing units and a 5.4 percent decline in total occupied housing units. The larger decline for owner-occupied units than for total occupied units yields the decrease in the homeownership rate. The fact that total occupied units decline suggest that homeowners insurance non-renewals do not simply move households from owner-occupied homes to rentals within a county, they also

shrink the county's total occupied housing base, a significant concern for its officials and community.

In summary, the results reported in Tables 6 through 8 show that homeowners insurance non-renewals have a broad range of significant adverse effects on the local economy. They reduce housing values, depress retail spending, and lower homeownership. These findings are consistent with insurance availability being critical for housing-market stability, household financial resilience, and the overall economy, a growing concern for policymakers as non-renewals become increasingly frequent in a broadening set of areas in the country.

6.4. Foreclosure Channel

Because foreclosure generally has profound adverse effects on households, we expect it is a key channel through which non-renewals of homeowners insurance policies affect the local economy. To test this expectation, we expand equation (4), with the various dependent variables examined in Tables 6 to 8, to include *HighForeclosure*, an indicator for counties with above-median foreclosure rates during the year, both separately and interacted with *NonrenewalRate*. To avoid the forbidden regression problem, we again instrument for the interactive variables using the share-shift instrument times *HighForeclosure* as the instrument.

Table 9 presents the estimations. Panel A examines housing prices, as in Table 6. The coefficient on the instrumented interaction of *HighForeclosure* and *NonrenewalRate* is significantly negative across the housing-price distribution, consistent with non-renewals depressing housing values more in counties with higher foreclosures. The coefficient is most negative for the value of low-tier housing, consistent with the owners of this housing being most financially constrained and prone to forced sales and other actions that depress the value of the housing.

Panel B examines retail sales, as in Table 7. The coefficient on the instrumented interaction of *HighForeclosure* and *NonrenewalRate* is significantly negative only for durable-goods sales. This finding is consistent with discussion in Section 6.2; the effect of foreclosure is larger for durable goods, because a substantial portion of these goods are items most likely to be purchased by homeowners.

Panel C examines homeownership, as in Table 8. The coefficient on the instrumented interaction of *HighForeclosure* and *NonrenewalRate* is at least weakly significantly negative in all three columns, consistent with foreclosure increasing the effects of non-renewals on homeownership.

In summary, Table 9 supports foreclosure as a key channel by which homeowners insurance non-renewals adversely affect the local economic variables examined in Tables 6-8. Non-renewals have larger adverse effects on housing prices, durable goods sales, and homeownership in counties for which foreclosures are higher.

7. Robustness

A concern about our shift-share-style instrument is that the instrument may capture county climate-disaster damages rather than just insurers' reduced provision of homeowners insurance coverage in the county. Specifically, although the shift component of the instrument is based on out-of-state climate-disaster damages, county climate-disaster damages may be correlated with out-of-state damages. To the extent that this is the case, and if county climate-disaster damages have direct effects foreclosures and local economic outcomes, then the exclusion restriction assumption for the instrument would be violated.

To address this concern, Table 10 reports re-estimates the main models in Tables 4 (foreclosures), 6 (housing values), 7 (retail sales), and 8 (homeownership) including the natural

logarithm of climate-disaster damages scaled by GDP for the county-year an additional control variable. The dependent variables are the one-year-ahead foreclosure rate in column (1), the natural logarithm of housing values in column (2), the natural logarithm of total retail sales in column (3), and the homeownership rate in column (4). The previously discussed results for these models remain robust to the inclusion of this control variable, which yields coefficients with similar magnitudes. The first-stage Kleibergen-Paap F statistics remain well above the Stock-Yogo critical value.

8. Conclusion

We examine whether insurers' non-renewal of homeowners insurance policies constitutes a channel through which climate change adversely affects U.S. households and local economies. Using national county-year non-renewals reported by the U.S. Senate Budget Committee covering 65 percent of the U.S. homeowners insurance market from 2018 to 2023, we trace this channel from climate-disaster damages to non-renewals of homeowners insurance policies, to home foreclosures, and finally to local housing values, retail sales, and homeownership. Our evidence indicates that the non-renewals constitute an economically meaningful channel for the adverse effects of climate change, that this channel works through the contractual requirement that mortgaged homes carry hazard insurance, and that most of the adverse effects of non-renewals are amplified by foreclosure.

We conduct cross-sectional analyses that show that the effects of homeowners insurance non-renewals are stronger in settings where we expect them to be stronger. Specifically, we find that higher climate damages in the county and the remainder of the state lead to higher non-renewals in states with less well capitalized insurers, homeowners insurance rates subject to regulatory prior approval, and higher uncertainty about climate-disaster damages, as well as in

counties with more socially vulnerable populations. We further find that higher non-renewals lead to higher foreclosures when there are fewer alternative providers of homeowners insurance in the state, there is no insurer of last resort in the state, there is non-judicial foreclosure in the state, and the county population is more socially vulnerable. We find that the adverse consequences of non-renewals for housing values are concentrated in the bottom tier of the distribution of housing values, that the adverse consequences of non-renewals for sales of durable goods, but not non-durable goods, are stronger when foreclosures are higher.

Our findings have implications beyond the specific setting we study. These findings suggest that climate adaptation in insurance markets depends not only on the premiums charged for coverage, but also on the willingness of insurers to continue offering coverage. Given ongoing climate change and more frequent climate disasters, homeowners insurance non-renewals are likely to become an increasingly important channel through which climate risk affects household financial stability, local housing markets, and social inequality.

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Appendix A. Variable Definitions

Variable	Definition	Data Source
<i>Outcome variables</i>		
NonrenewalRate	The number of non-renewed homeowners insurance policies divided by the total number of those policies in force prior to non-renewals in the county-year, in percentage points	Senate Budget Committee
Foreclosure	The sum of the numbers of notices of trustee sale (NTS) and notices of foreclosure sales (NFS) divided by the number of housing units with a mortgage in the county-year, in percentage points	ATTOM; Census ACS
Log(HousePrice)	The natural logarithms of the average values of four Zillow Home Value Indices (ZHVI) for the county-year: all single-family, bottom-tier (5 th -35 th percentile), middle-tier (35 th -65 th percentile), and top-tier (67-100 percentile) homes	Zillow
Log(RetailSales)	The natural logarithms of total, durable goods, and non-durable goods sales in retail establishments in the county-year. Total retail sales include sales in SIC codes 52–59. Durable-goods sales include only SIC codes 52 (building/hardware), 55 (motor vehicles), 57 (home furniture), 5940–5949 (shopping goods). Non-durable goods sales include only SIC codes 54 (food), 56 (apparel), 5912/5921/5331 (drugs/liquor/variety), 5540–5549 (gas stations), 5980–5989 (fuel dealers), 5812 (eating), 5813 (drinking).	Data Axle
Homeownership	The number of owner-occupied housing units divided by the number of occupied units in the county-year, in percentage points.	Census ACS
Log(Owners)	The natural logarithm of owner-occupied housing units in the county-year	Census ACS
Log(TotalOcc)	The natural logarithm of occupied housing units in the county-year	Census ACS
<i>Climate-damage variables</i>		
LocalDamage_raw	Average annual county-level climate-disaster damages (excluding crop damages and damages due to earthquakes, floods, tsunamis, and volcanos) scaled by real GDP in the county across the three years ending in the current year, multiplied by 100.	SHELDUS; BEA
StateOtherDamage_raw	Average annual climate-disaster damages to property (excluding crop damages and damages due to earthquakes, floods, tsunamis, and volcanos) scaled by real GDP in the rest of the state (i.e., excluding the focal county) across the three years ending in the current year, multiplied by 100.	SHELDUS; BEA
LocalDamage	Within-year quintile rank of LocalDamage_raw, rescaled to range from 0 to 1.	SHELDUS; BEA
StateOtherDamage	Within-year quintile rank of StateOtherDamage_raw, rescaled to range from 0 to 1.	SHELDUS; BEA

Instrument variable

IV	Shift-share-style instrumental variable that equals the product of the predetermined percentile rank of homeowners-insurance policies per owner-occupied housing unit in the county at year 2018 (the share) and the natural logarithm of average out-of-state climate damages scaled by out-of-state real GDP over the three years ending in the current year (the shift).	NAIC; Census ACS; SHELDUS; BEA
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Control variables

BlackShare	Share of population that is Black or African American in the county-year.	Census ACS
Age20-44Share	Share of population aged 20 to 44 in the county-year.	Census ACS
Unemployment	Unemployment rate in percentage points in the county-year.	BLS
Log(IncomePC)	Natural logarithm of per capita personal income in the county-year.	BEA

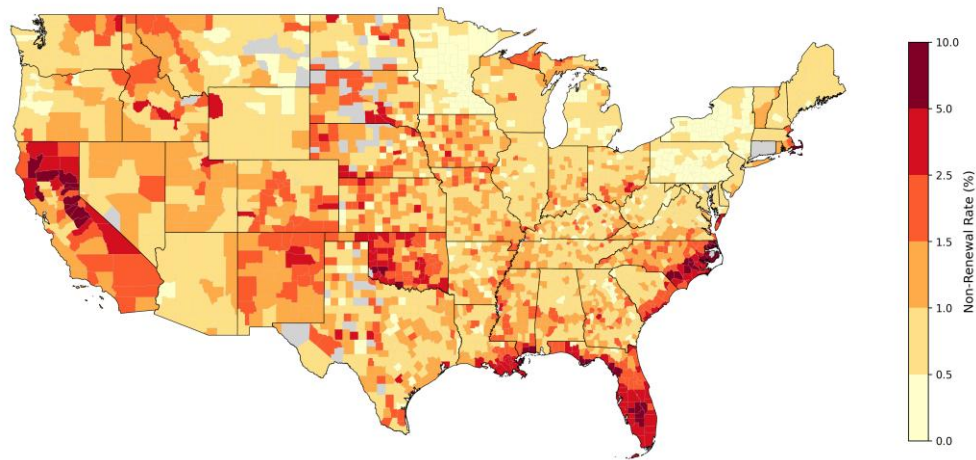
Cross-sectional moderating variables

WeakCapital	Indicator equal to 1 for state-years with below-median weighted-average insurer Risk-Based Capital (RBC) ratio, 0 otherwise.	NAIC
Frictions	Indicator equal to 1 for states with homeowners-insurance rate filings subject to prior approval regulation, 0 otherwise. This variable is constant over time.	NAIC Rate Filing Methods for P&C Insurance
Uncertainty	Within-year quintile ranks of the coefficient of variation of climate-disaster damages scaled by GDP for the state over the prior three years, rescaled to range from 0 to 1.	SHELDUS
SVI	CDC/ATSDR Social Vulnerability Index in the county-year, standardized to have a mean of zero and standard deviation of one.	CDC/ATSDR
#Insurers	Number of homeowners-insurance carriers operating in the state-year, standardized to have a mean of zero and standard deviation of one.	NAIC
HHI	Herfindahl-Hirschman Index of homeowners-insurance market concentration in the state-year, standardized to have a mean of zero and standard deviation of one.	NAIC
NoFairPlan	Indicator equal to 1 for states without a Fair Access to Insurance Requirements (FAIR) plan or other homeowners insurance provider of last resort, 0 otherwise.	State regulatory filings; FAIR plan registries
Nonjudicial	Indicator equal to 1 for states with non-judicial foreclosure procedures, 0 otherwise. This variable is constant over time.	State foreclosure-law summaries
HighForeclosure	Indicator equal to 1 for county-years with foreclosure rates above the within-year median, and 0 otherwise.	ATTOM; Census ACS

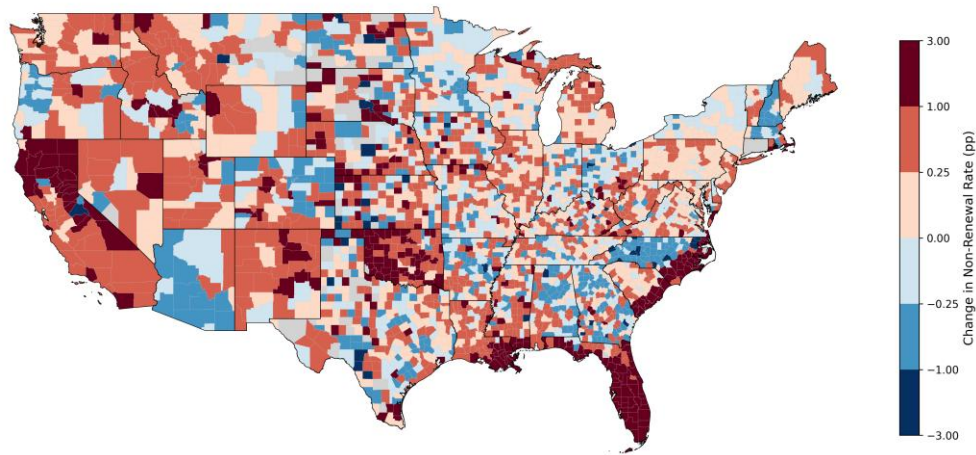
This Appendix contains definitions of the variables used in this study and the data sources for those variables.

Figure 1. Geographic Distribution of Homeowner-Insurance Non-Renewal Rates

Panel A: Non-Renewal Rate by County, 2023 (percent)



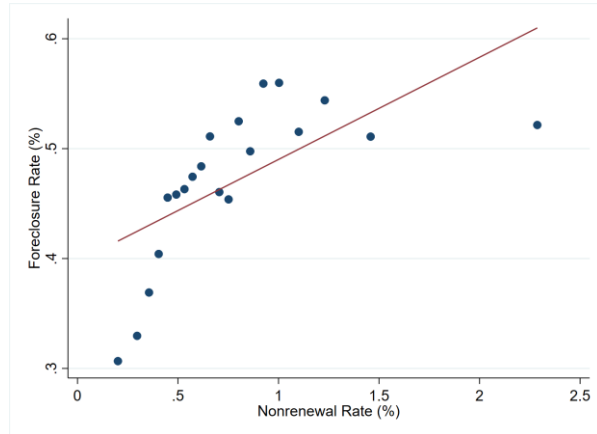
Panel B: Change in Non-Renewal Rate, 2018-2023 (percentage points)



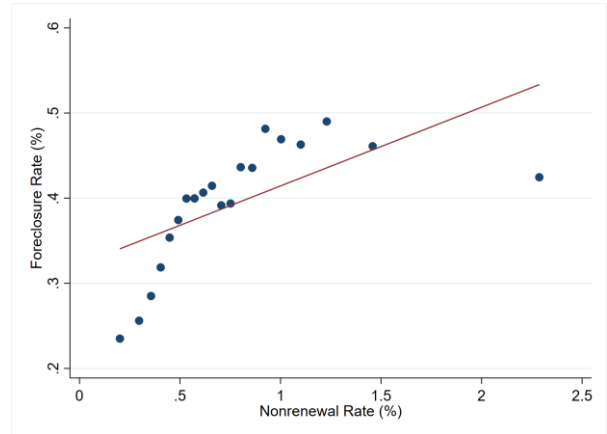
This figure depicts the level and changes in homeowner-insurance non-renewal rates by U.S. county. Panel A depicts the non-renewal rate in 2023 in percent and Panel B depicts the change in the non-renewal rate from 2018 to 2023 in percentage points. *Source:* Senate Budget Committee.

Figure 2. Binned Scatterplots of Levels and Changes in Non-Renewal Rates and Foreclosure Rates

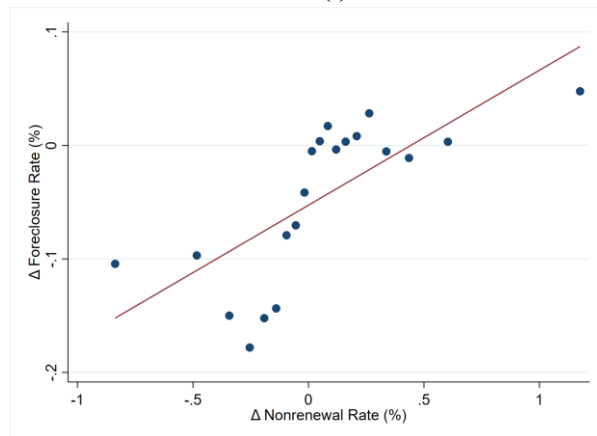
Panel A: Foreclosure rate (t) vs. Non-renewal rate



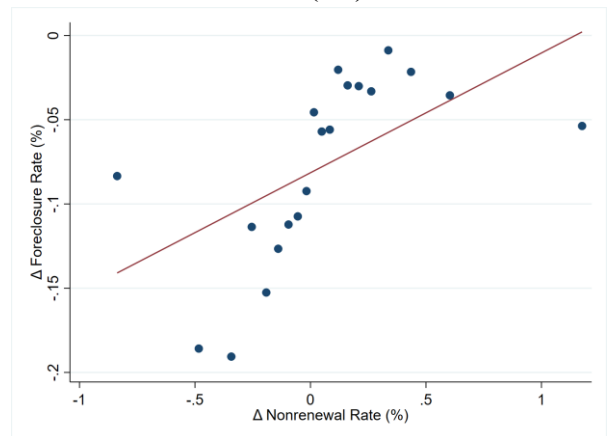
Panel B: Foreclosure rate (t+1) vs. Non-renewal rate



Panel C: Δ Foreclosure rate (t) vs. Δ Non-renewal rate



Panel D: Δ Foreclosure rate (t+1) vs. Δ Non-renewal rate



This figure presents binned scatterplots of the relationship between the county-year homeowner-insurance non-renewal rates and foreclosure rates, in levels (top row, Panels A and B) and in first differences (bottom row, Panels C and D). Panels A and C use the contemporaneous foreclosure rate (year t); Panels B and D use the one-year-ahead foreclosure rate (year t+1). The sample covers 2018-2023; All variables are winsorized at the 1st and 99th percentiles. Solid lines depict linear fits.

Table 1. Summary Statistics

Variable	N	Mean	SD	p25	Median	p75
<i>Climate-damage variables and controls</i>						
LocalDamage_raw	16,281	0.089	0.448	0.000	0.003	0.016
StateOtherDamage_raw	16,281	0.079	0.269	0.004	0.011	0.059
LocalDamage	16,281	0.500	0.354	0.250	0.500	0.750
StateOtherDamage	16,281	0.500	0.354	0.250	0.500	0.750
BlackShare	16,281	0.105	0.143	0.016	0.037	0.131
Age20-44Share	16,281	0.300	0.041	0.274	0.295	0.321
Unemployment	16,281	4.720	1.841	3.392	4.300	5.625
Log(IncomePC)	16,281	10.740	0.230	10.580	10.719	10.871
<i>Non-renewal, IV and foreclosure variables</i>						
NonrenewalRate	16,109	0.783	0.484	0.470	0.680	0.955
IV	16,109	-3.564	2.041	-5.321	-3.562	-1.798
Foreclosure (t)	16,109	0.475	0.610	0.081	0.289	0.625
Foreclosure (t+1)	16,109	0.399	0.496	0.056	0.253	0.537
<i>Housing variables</i>						
Log(HousePrice, overall)	15,759	12.208	0.504	11.848	12.156	12.523
Log(HousePrice, bottom tier)	15,759	11.675	0.597	11.239	11.636	12.075
Log(HousePrice, middle tier)	15,759	12.202	0.495	11.848	12.154	12.516
Log(HousePrice, top tier)	15,759	12.732	0.447	12.415	12.679	12.986
<i>Retail sales variables</i>						
Log(RetailSales, total)	16,096	13.060	1.586	11.875	12.915	14.115
Log(RetailSales, durable)	16,096	11.909	1.813	10.570	11.793	13.182
Log(RetailSales, non-durable)	16,096	12.362	1.455	11.282	12.190	13.291
<i>Homeownership variables</i>						
Homeownership	16,109	72.398	7.961	67.970	73.527	78.002
Log(Owners)	16,109	9.328	1.225	8.446	9.153	10.051
Log(TotalOcc)	16,109	9.659	1.268	8.739	9.456	10.404

This table reports summary statistics for the main model variables. All variables are winsorized at the 1st and 99th percentiles. Definitions of all variables are listed in Appendix A.

Table 2. Effect of Climate-Disaster Damages on Homeowners Insurance Non-Renewal Rates

Dep. Var. =	NonrenewalRate		
	(1)	(2)	(3)
LocalDamage	0.038** (0.017)		0.033* (0.017)
StateOtherDamage		0.040** (0.017)	0.035** (0.017)
BlackShare	-5.958*** (1.464)	-5.921*** (1.459)	-5.957*** (1.458)
Age20-44Share	-4.948*** (1.530)	-4.897*** (1.531)	-4.917*** (1.531)
Unemployment	-0.003 (0.005)	-0.002 (0.005)	-0.002 (0.005)
Log(IncomePC)	0.004 (0.134)	-0.012 (0.135)	-0.003 (0.135)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	16,281	16,281	16,281
Adj. R-squared	0.487	0.487	0.487

This table reports OLS estimates of the impact of recent climate-disaster damages in the county and the remainder of the state on the homeowner insurance non-renewal rate in the county during the year. The dependent variable is *NonrenewalRate*, the homeowner-insurance non-renewal rate in the county-year. The explanatory variables of interest, *LocalDamage* and *StateOtherDamage*, are the within-year quintile ranks of the county's and the rest-of-state's three-year-average climate-related property damage scaled by real GDP during the year. Control variables are lagged county characteristics, including the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita in the county-year. All specifications include county and year fixed effects. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively. See Appendix A for variable definitions.

Table 3. Tests of Cross-Sectional Modifiers of the Effect of Climate-Disaster Damages on Homeowners Insurance Non-Renewal Rates

Dep. Var. =	Nonrenewal Rate			
	(1)	(2)	(3)	(4)
LocalDamage	0.011 (0.017)	0.026 (0.018)	-0.001 (0.021)	-0.034 (0.031)
StateOtherDamage	-0.008 (0.019)	-0.006 (0.016)	-0.296*** (0.034)	-0.178*** (0.035)
WeakCapital	-0.056*** (0.018)			
LocalDamage × WeakCapital	0.043** (0.021)			
StateOtherDamage × WeakCapital	0.094*** (0.020)			
LocalDamage × Frictions		0.027 (0.045)		
StateOtherDamage × Frictions		0.137*** (0.049)		
Uncertainty			-0.203*** (0.026)	
LocalDamage × Uncertainty			0.097*** (0.033)	
StateOtherDamage × Uncertainty			0.477*** (0.046)	
SVI				-0.038 (0.086)
LocalDamage × SVI				0.125** (0.060)
StateOtherDamage × SVI				0.373*** (0.060)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	16,281	16,281	16,281	16,281
Adj. R-squared	0.488	0.487	0.500	0.490

This table reports OLS estimations of the effects of climate-disaster damages in the county and the remainder of the state on homeowner insurance non-renewal rate, including four cross-sectional moderating variables interacted with the climate damage variables and included separately if they vary over time. *WeakCapital* equals 1 for states for which the insurers serving the state have below-median weighted average insurer risk-based capital ratio in the prior year and 0 otherwise. *Frictions* equals 1 for states with high insurance regulatory frictions in the pricing of homeowners insurance; this indicator does not vary over time. *Uncertainty* is the within-year quintile ranks of the coefficient of variation of the climate-disaster damages scaled by GDP for the state over the prior three years, rescaled to range from 0 to 1. *SVI* is the county-level Social Vulnerability Index. Control variables are lagged county demographic and economic characteristics, including the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. All specifications include county and year fixed effects. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively. See Appendix A for variable definitions.

Table 4. Effects of Homeowners Insurance Non-Renewal Rates on Foreclosure Rates

Dep. Var. =	(1) Foreclosure (t)	(2) Foreclosure (t+1)
NonrenewalRate	0.917*** (0.171)	0.711*** (0.129)
BlackShare	3.468 (2.540)	1.550 (1.843)
Age20-44Share	7.735*** (2.421)	9.042*** (1.820)
Unemployment	-0.010 (0.007)	0.014** (0.005)
Log(IncomePC)	0.275 (0.172)	0.178 (0.131)
County FE	Yes	Yes
Year FE	Yes	Yes
Observations	16,109	16,109
Kleibergen-Paap F	58.44	58.44
Stock-Yogo 10%	16.38	16.38

This table reports two-stage least-squares (2SLS) estimations of the effects of homeowner-insurance non-renewal rates on the foreclosure rate in the county-year. The dependent variable *Foreclosure* is the foreclosure rate in the county-year in year t in column (1) and year t+1 in column (2). The explanatory variable of interest, *NonrenewalRate*, is the homeowners insurance non-renewal rate in the county-year, which is instrumented using the shift-share-style instrument defined in Appendix A. Control variables are lagged county characteristics, including the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita in the county-year. All specifications include county and year fixed effects. The Kleibergen–Paap rk Wald F statistic and the Stock–Yogo (2005) critical value for 10% maximal IV size are reported at the bottom of the table. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively. See Appendix A for variable definitions.

Table 5. Tests of Cross-Sectional Modifiers of the Effects of Homeowners Insurance Non-Renewal Rates on Foreclosure Rates

Panel A: Foreclosure (t)					
	(1) #Insurers	(2) HHI	(3) NoFairPlan	(4) Nonjudicial	(5) SVI
NonrenewalRate	1.295*** (0.252)	0.963*** (0.182)	0.872*** (0.187)	0.159 (0.145)	0.343 (0.239)
NonrenewalRate × Partition	-1.540*** (0.435)	0.449*** (0.106)	0.823*** (0.167)	1.313*** (0.148)	0.514*** (0.183)
Partition	1.621*** (0.359)	-0.683*** (0.088)			-0.613*** (0.141)
Controls	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	16,109	16,109	16,109	16,109	16,109
Kleibergen-Paap F	10.03	23.66	30.02	28.55	11.08
Stock-Yogo 10%	7.03	7.03	7.03	7.03	7.03

Panel B: Foreclosure (t+1)					
	(1) #Insurers	(2) HHI	(3) NoFairPlan	(4) Nonjudicial	(5) SVI
NonrenewalRate	1.037*** (0.208)	0.772*** (0.138)	0.679*** (0.141)	0.034 (0.115)	0.174 (0.181)
NonrenewalRate × Partition	-1.277*** (0.349)	0.226*** (0.087)	0.589*** (0.126)	1.174*** (0.110)	0.491*** (0.141)
Partition	1.330*** (0.290)	-0.493*** (0.068)			-0.506*** (0.109)
Controls	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	16,109	16,109	16,109	16,109	16,109
Kleibergen-Paap F	10.03	23.66	30.02	28.55	11.08
Stock-Yogo 10%	7.03	7.03	7.03	7.03	7.03

This table reports two-stage least squares (2SLS) estimations of the relation between homeowners insurance non-renewal rates and foreclosure rates during the county-year, including five cross-sectional moderating variables interacted with non-renewal rates and separately if they vary over time. The dependent variable is the foreclosure rate in the county in year t in Panel A and in year $t+1$ in Panel B. The explanatory variables of interest are the interactions of *NonrenewalRate* with each partitioning variable. To avoid the forbidden regression problem in interactive 2SLS models, we instrument for these interactive variables using the interactions of the shift-share-style instrument described in Appendix A with each partitioning variable. The partitioning variables are the number of homeowners-insurance carriers operating in the state (*#Insurers*) in column (1), the Herfindahl-Hirschman Index of homeowners-insurance market concentration in the state (*HHI*) in column (2), an indicator for states without FAIR plans or other insurance providers of last resort (*NoFairPlan*) in column (3), an indicator for states with non-judicial foreclosure procedures (*Nonjudicial*) in column (4), and the Social Vulnerability Index for the county (*SVI*) in column (5). The lagged county characteristic control variables include the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. All specifications include county and year fixed effects. The Kleibergen–Paap rk Wald F statistic and the Stock–Yogo (2005) critical value for 10% maximal IV size are reported at the bottom of the table. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively. See Appendix A for variable definitions.

Table 6. Effect of Homeowners Insurance Non-Renewal Rates on Housing Prices

Dep. Var. =	Log(HousePrice)			
	(1) Overall	(2) Bottom tier	(3) Middle tier	(4) Top tier
NonrenewalRate	-0.083** (0.034)	-0.165*** (0.046)	-0.087** (0.034)	-0.095*** (0.028)
BlackShare	-0.706* (0.420)	-0.493 (0.569)	-0.735* (0.422)	-1.029*** (0.379)
Age20-44Share	2.455*** (0.412)	2.044*** (0.560)	2.456*** (0.412)	2.141*** (0.384)
Unemployment	-0.005*** (0.001)	-0.004** (0.002)	-0.005*** (0.001)	-0.002* (0.001)
Log(IncomePC)	0.147*** (0.032)	0.083* (0.043)	0.144*** (0.032)	0.126*** (0.030)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	15,759	15,759	15,759	15,759
Kleibergen-Paap F	56.37	56.37	56.37	56.37
Stock-Yogo 10%	16.38	16.38	16.38	16.38

This table reports two-stage least-squares (2SLS) estimations of the effect of homeowner insurance non-renewal rates on the housing prices in the county-year. The dependent variable $\text{Log}(\text{HousePrice})$ is the natural logarithm of four Zillow Home Value Indices (ZHVI) in year $t+1$: the overall single-family ZHVI in column (1), and the bottom (5th-35th percentile), middle (35th-65th percentile), and top (65th-95th percentile) tier ZHVIs in columns (2)–(4), respectively. The explanatory variable of interest, *NonrenewalRate*, is the homeowners insurance non-renewal rate in the county year, which is instrumented using the share-shift-style instrument described in Appendix A. Control variables are lagged county characteristics, including the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. All specifications include county and year fixed effects. The Kleibergen–Paap rk Wald F statistic and the Stock–Yogo (2005) critical value for 10% maximal IV size are reported at the bottom of the table. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively. See Appendix A for variable definitions.

Table 7. Impact of Homeowners Insurance Non-Renewal Rates on Retail Sales

Dep. Var. =	Log(RetailSales)		
	(1) Total	(2) Durable	(3) Non-durable
NonrenewalRate	-0.086** (0.041)	-0.106* (0.059)	-0.129*** (0.050)
BlackShare	-1.120* (0.648)	-0.566 (0.791)	-0.973 (0.699)
Age20-44Share	-0.081 (0.736)	-0.754 (0.808)	0.195 (0.751)
Unemployment	-0.002 (0.002)	0.000 (0.002)	-0.006** (0.003)
Log(IncomePC)	0.070 (0.061)	-0.015 (0.064)	0.045 (0.067)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	16,096	16,096	16,096
Kleibergen-Paap F	58.24	58.24	58.24
Stock-Yogo 10%	16.38	16.38	16.38

This table reports two-stage least-squares (2SLS) estimations of the effect of the homeowners insurance non-renewal rate on retail sales in the county-year. The dependent variable, $\text{Log}(\text{RetailSales})$, is the natural logarithm of three measures of retail establishment sales in year $t+1$: total retail sales in column (1); durable-goods sales in column (2); and non-durable-goods sales in column (3). The main explanatory variable of interest, *NonrenewalRate*, is the homeowners insurance non-renewal rate in the county-year, which is instrumented using the share-shift-style instrumental variable described in Appendix A. Control variables are lagged county characteristics, including the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. All specifications include county and year fixed effects. The Kleibergen–Paap rk Wald F statistic and the Stock–Yogo (2005) critical value for 10% maximal IV size are reported at the bottom of the table. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively. See Appendix A for variable definitions.

Table 8. Impact of Homeowners Insurance Non-Renewal Rates on Homeownership Rates

Dep. Var. =	(1) Homeownership	(2) Log(Owners)	(3) Log(TotalOcc)
NonrenewalRate	-1.591** (0.692)	-0.138*** (0.025)	-0.112*** (0.020)
BlackShare	-25.220*** (9.052)	-0.364 (0.350)	-0.006 (0.290)
Age20-44Share	10.978 (9.134)	0.227 (0.361)	0.137 (0.301)
Unemployment	-0.052* (0.026)	0.003*** (0.001)	0.003*** (0.001)
Log(IncomePC)	-0.183 (0.763)	-0.075*** (0.027)	-0.077*** (0.022)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	16,109	16,109	16,109
Kleibergen-Paap F	58.44	58.44	58.44
Stock-Yogo 10%	16.38	16.38	16.38

This table reports two-stage least-squares (2SLS) estimations of the effect of homeowners insurance non-renewal rates on three homeownership outcomes from the Census ACS estimates in the county in the following year: the homeownership rate (*Homeownership*) in column (1), the natural logarithm of the number of owner-occupied housing units (*Log(Owners)*) in column (2), and the natural logarithm of the number of total occupied units (*Log(TotalOcc)*) in column (3). The main independent variable, *NonrenewalRate*, is the homeowners insurance non-renewal rate in the county-year, which is instrumented using the share-shift-style instrumental variable described in Appendix A. Control variables are lagged county characteristics, including the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. All specifications include county and year fixed effects. The Kleibergen–Paap rk Wald F statistic and the Stock–Yogo (2005) critical value for 10% maximal IV size are reported at the bottom of the table. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively. See Appendix A for variable definitions.

Table 9. Tests of Foreclosure Rates as a Moderating Variable for the Effects of Homeowners Insurance Non-renewal Rates on Local Economic Variables

Panel A: Log(HousePrice) (t+1)

	(1) Single-family	(2) Bottom tier	(3) Middle tier	(4) Top tier
NonrenewalRate	-0.046 (0.028)	-0.105*** (0.039)	-0.051* (0.028)	-0.049* (0.025)
HighForeclosure	0.049* (0.026)	0.075** (0.035)	0.047* (0.026)	0.065*** (0.024)
NonrenewalRate $\times$ HighForeclosure	-0.074** (0.034)	-0.118** (0.047)	-0.072** (0.035)	-0.096*** (0.031)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	15,759	15,759	15,759	15,759
Kleibergen-Paap F	25.03	25.03	25.03	25.03
Stock-Yogo 10%	7.03	7.03	7.03	7.03

Panel B: Log(RetailSales) (t+1)

	(1) Total	(2) Durable	(3) Non-durable
NonrenewalRate	-0.094** (0.039)	-0.025 (0.055)	-0.160*** (0.048)
HighForeclosure	-0.018 (0.051)	0.109* (0.057)	-0.053 (0.059)
NonrenewalRate $\times$ HighForeclosure	0.021 (0.068)	-0.164** (0.077)	0.070 (0.079)
Controls	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	16,096	16,096	16,096
Kleibergen-Paap F	27.24	27.24	27.24
Stock-Yogo 10%	7.03	7.03	7.03

Panel C: Homeownership outcomes (t+1)

	(1) Homeownership	(2) Log(Owners)	(3) Log(TotalOcc)
NonrenewalRate	-0.978 (0.603)	-0.101*** (0.023)	-0.085*** (0.019)
HighForeclosure	0.858* (0.505)	0.040** (0.020)	0.027 (0.017)
NonrenewalRate $\times$ HighForeclosure	-1.267* (0.667)	-0.068*** (0.026)	-0.048** (0.022)
Controls	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	16,109	16,109	16,109
Kleibergen-Paap F	27.34	27.34	27.34
Stock-Yogo 10%	7.03	7.03	7.03

This table reports IV (2SLS) estimations of the effect of homeowners insurance non-renewal rates on following-year local-economic variables in the county, including an indicator for high foreclosures interacted with non-

renewal rates and separately. *HighForeclosure* equals 1 for counties with foreclosure rates above the within-year median, and 0 otherwise. The explanatory variables of interest are *NonrenewalRate* and $NonrenewalRate \times HighForeclosure$, which are instrumented by the share-shift-style indicator *IV* and by $IV \times HighForeclosure$, respectively. The dependent variables are the natural logarithms of Zillow's housing values (single-family / bottom tier / middle tier / top tier) in Panel A; the natural logarithms of retail sales (total sales/ durable sales/ non-durable sales) in Panel B; the homeownership rate, natural logarithm of the number of owner-occupied housing units, and the natural logarithm of the number of total occupied units in Panel C. Controls include lagged county characteristics, including the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. All specifications include county and year fixed effects. The Kleibergen–Paap rk Wald F statistic and the Stock–Yogo (2005) critical value for 10% maximal IV size are reported at the bottom of the table. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively. See Appendix A for variable definitions.

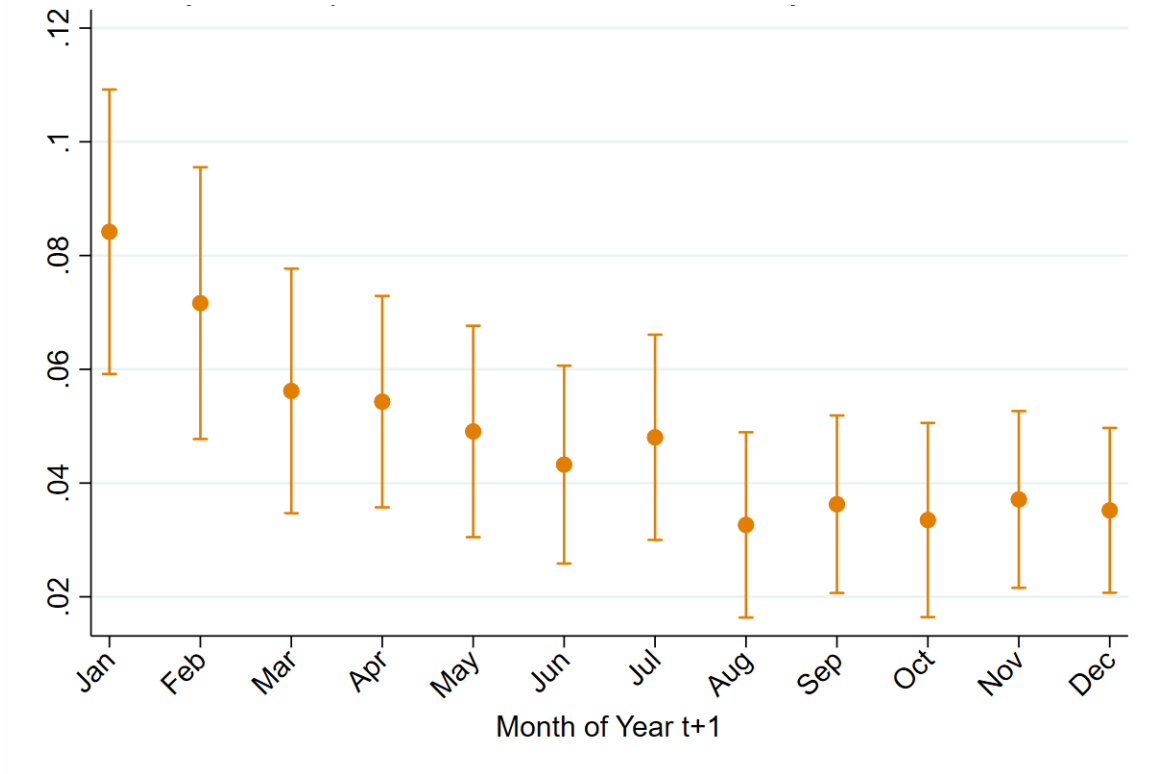
Table 10. Robustness to Controlling for County Climate-Disaster Damages

Dep. Var. =	(1) Foreclosure	(2) Log(HousePrice)	(3) Log(RetailSales)	(4) Homeownership
NonrenewalRate	0.665*** (0.115)	-0.091*** (0.031)	-0.076** (0.038)	-1.519** (0.641)
Log(LocalDamage)	-4.122** (1.833)	-0.677* (0.374)	0.904* (0.462)	6.368 (7.670)
BlackShare	1.562 (1.799)	-0.702* (0.420)	-1.123* (0.644)	-25.238*** (9.035)
Age20-44Share	8.803*** (1.733)	2.413*** (0.408)	-0.028 (0.720)	11.347 (9.012)
Unemployment	0.014*** (0.005)	-0.005*** (0.001)	-0.002 (0.002)	-0.052** (0.026)
Log(IncomePC)	0.165 (0.126)	0.144*** (0.032)	0.073 (0.061)	-0.163 (0.760)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	16,109	15,759	16,096	16,109
Kleibergen-Paap F	67.16	64.47	66.96	67.16
Stock-Yogo 10%	16.38	16.38	16.38	16.38

This table reports two-stage least-squares (2SLS) estimations of the main models in Tables (4) and (6)-(8) including local climate damage as an additional control. *Log(LocalDamage)* is the natural logarithm of one plus the three-year average climate-disaster damage scaled by local real GDP in the county. The dependent variables are the foreclosure rate (*Foreclosure*) in column (1), the natural logarithm of single-family Zillow Home Value Index (*Log(HousePrice)*) in column (2), the natural logarithm of retail establishment sales (*Log(RetailSales)*) in column (3), and the homeownership rate (*Homeownership*) in column (4). All outcome variables are measured in year $t+1$. The explanatory variable of interest, *NonrenewalRate*, is the homeowners insurance non-renewal rate, which is instrumented using the shift-share-style instrumental variable described in Appendix A. Control variables are lagged county characteristics, including the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. All specifications include county and year fixed effects. The Kleibergen–Paap rk Wald F statistic and the Stock–Yogo (2005) critical value for 10% maximal IV size are reported at the bottom of the table. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level, respectively. See Appendix A for variable definitions.

Internet Appendix

Figure IA1. Dynamic Impact of Non-Renewals on Monthly Foreclosure Rate



This figure plots two-stage least-squares (2SLS) estimates of the dynamic impact of the county-year homeowner-insurance non-renewal rate on the monthly foreclosure rate over the twelve months of year $t+1$. Each marker corresponds to a separate regression of that month's foreclosure rate on *NonrenewalRate*, instrumented using the instrument *IV* described in Appendix A. Vertical bars are 95% confidence intervals, with standard errors clustered at the county level. All specifications include the lagged county characteristic controls as well as county and year fixed effects. See Appendix A for variable definitions.

Table IA1. Annual Means and Medians of Non-Renewal and Foreclosure Rates

Variable	2018	2019	2020	2021	2022	2023
Mean						
NonrenewalRate	0.811	0.715	0.533	0.736	0.929	1.126
Foreclosure	0.746	0.738	0.385	0.210	0.354	0.507
Median						
NonrenewalRate	0.744	0.630	0.446	0.664	0.761	0.896
Foreclosure	0.458	0.420	0.208	0.142	0.265	0.384

This table reports annual means and medians of the homeowner-insurance non-renewal rate (*NonrenewalRate*) and the foreclosure rate (*Foreclosure*) for the full county-year panel over 2018-2023. Both variables are in percentage points. See Appendix A for detailed variable definitions.

Table IA2. First-Stage IV Diagnostics

Dep. Var. =	NonrenewalRate	
	(1)	(2)
IV	0.455*** (0.056)	0.424*** (0.055)
BlackShare		-5.386*** (1.445)
Age20-44Share		-4.748*** (1.641)
Unemployment		-0.005 (0.005)
Log(IncomePC)		-0.061 (0.136)
County FE	Yes	Yes
Year FE	Yes	Yes
Observations	16,109	16,109
Adj. R-squared	0.489	0.491
Kleibergen-Paap F	67.12	58.44
Stock-Yogo 10% crit. val.	16.38	16.38

This table reports the first-stage OLS relationship between the instrument *IV* and the endogenous regressor *NonrenewalRate*. Both columns use the estimation sample of the baseline IV regression in Table 4 (Foreclosure at *t*). Column (1) omits controls; column (2) adds the standard lagged controls, including the share of Black or African American residents, the share of residents aged 20 to 44, the unemployment rate, and the natural logarithm of income per capita. All specifications include county and year fixed effects, with standard errors clustered at the county level (in parentheses). The Kleibergen-Paap rk Wald F statistic (reported at the bottom) is computed from the corresponding 2SLS regression; values above the Stock-Yogo (2005) critical value for 10% maximal IV size (16.38, one endogenous regressor with one instrument) indicate a strong instrument. Standard errors clustered at the county level are in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% level. See Appendix A for detailed variable definitions.